Machine Learning: Midterm

Name: _____

1. Evaluating classifiers (20 points)

A university wants to use an AI-text detector to check student essays. To test it, the detector is run on 400 essays whose authorship is known: 60 were written with AI and 340 by students. Of the 400 essays, the detector flags 50 as written with AI. Of these 50 flagged essays, 45 really were written with AI. Treat "written with AI" as the positive class.

1.1 [4 points] Write down the confusion matrix.

	flagged	not flagged
written with AI		
written by students		

1.2 [2 points] What are precision and recall for this detector?

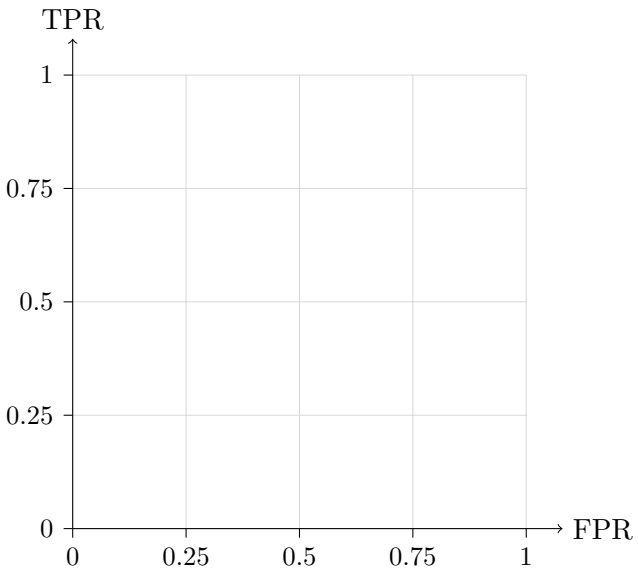
1.3 [2 points] Do we care more about improving precision or recall for this problem, and why?

The detector also gives each essay a score between 0 and 1; a higher score means "more likely written with AI". Here are 8 essays, 4 written with AI and 4 by students:

essay	1	2	3	4	5	6	7	8
score	0.95	0.85	0.70	0.60	0.50	0.40	0.30	0.10
written with AI?	yes	yes	no	yes	no	yes	no	no

We say the detector flags an essay as AI-written if its score is over t.

1.4 [10 points] Draw the ROC curve.



1.5 [2 points] Compute the AUC.

2. Support vector machines (20 points)

2.1 [10 points] A soft-margin SVM returned $\boldsymbol{w} = (6, 8)$, $b = -20$.

point	A	B	C	D	E
x_1	3	1	1.5	1.25	3
x_2	1	1	1.5	1.5	0.5
label	+	−	+	−	−

(a) Compute $f(\boldsymbol{x})$ and the predicted class of each point.

(b) What is the width of the margin?

(c) Compute each point's distance to the decision boundary.

(d) Which points are support vectors? Which point is on the margin, which is inside it, and which is misclassified?

2.2 [4 points] The soft-margin SVM solves

$$\min_{\boldsymbol{w}, b, \boldsymbol{\xi}} \ \tfrac{1}{2}\|\boldsymbol{w}\|^2 + C\sum_i \xi_i \quad \text{subject to} \quad y_i(\boldsymbol{w}\cdot\boldsymbol{x}_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0.$$

(a) What does the slack ξ_i measure, and why do we need it?

(b) Where does a training point lie, relative to the margin and the decision boundary, if $\xi_i = 0$? If $0 < \xi_i < 1$? If $\xi_i > 1$?

2.3 [4 points] Two soft-margin SVMs are trained on the same data, one with $C = 0.01$ and one with $C = 100$.

(a) Which has the wider margin, and why?

(b) Which has more support vectors, and why?

(c) Which is more likely to overfit, and why?

(d) How would you choose C in practice, and why?

2.4 [2 points] You remove a training point that is not a support vector and retrain the SVM. What changes? Why?

3. Nonlinear regression (20 points)

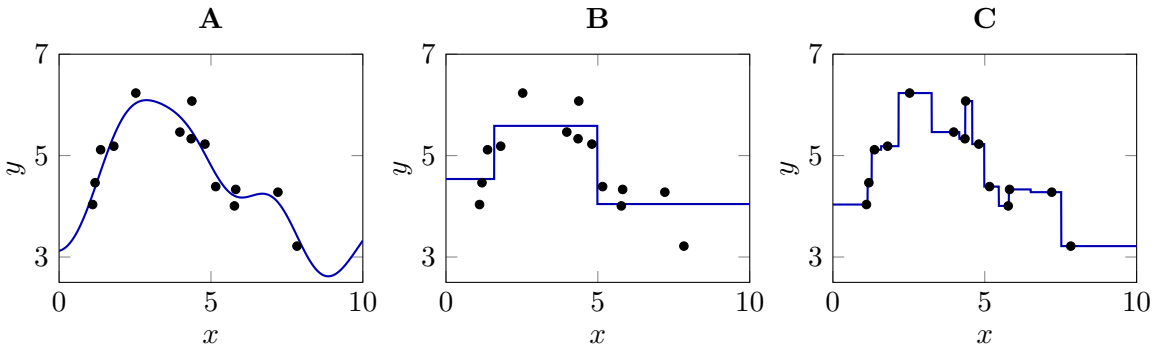
3.1 [4 points] kNN regression is trained on six points:

x	1	2	4	5	7	8
y	3	6	4	8	10	12

Predict y at $x = 4.4$ using (a) $k = 1$ and (b) $k = 3$.

3.2 [6 points] For each of a regression tree, kNN regression and SVR, name a hyperparameter that controls how closely the model follows the training data. Which setting of it makes overfitting more likely?

3.3 [5 points] The figure shows three models fitted to the same data. Which panel shows a regression tree, which shows kNN with $k = 1$, and which shows SVR with a kernel? Explain each.



3.4 [5 points] We use kNN to predict house prices from floor area (500 to 5,000 square feet) and number of bedrooms (1 to 6). Why might kNN do badly on these raw features? What would you do?